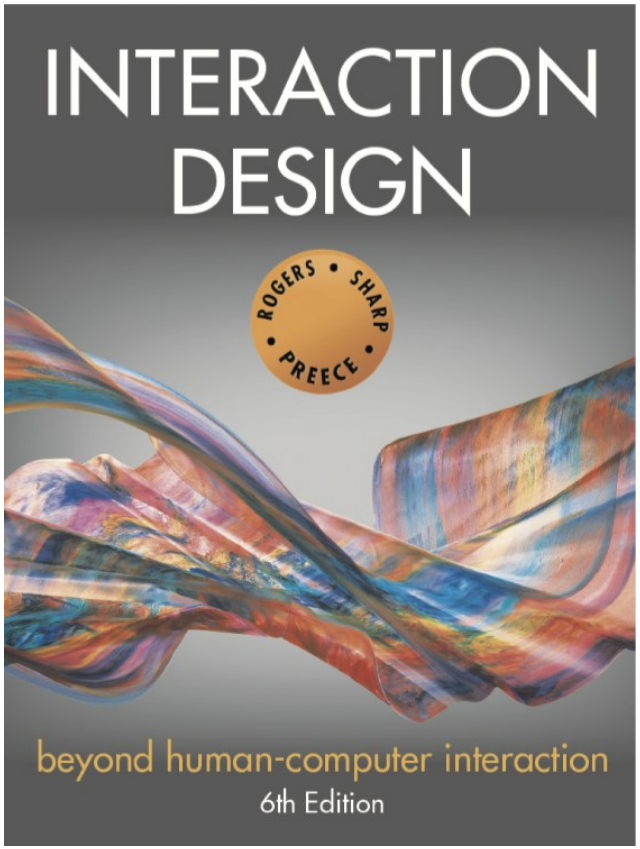


Chapter 7
INTERFACES

Overview

- Provide an overview of the diversity of interfaces
- Highlight the main design considerations for each of the interfaces
- Discuss what is meant by a natural user interface
- Consider which interface is best for a given application or activity

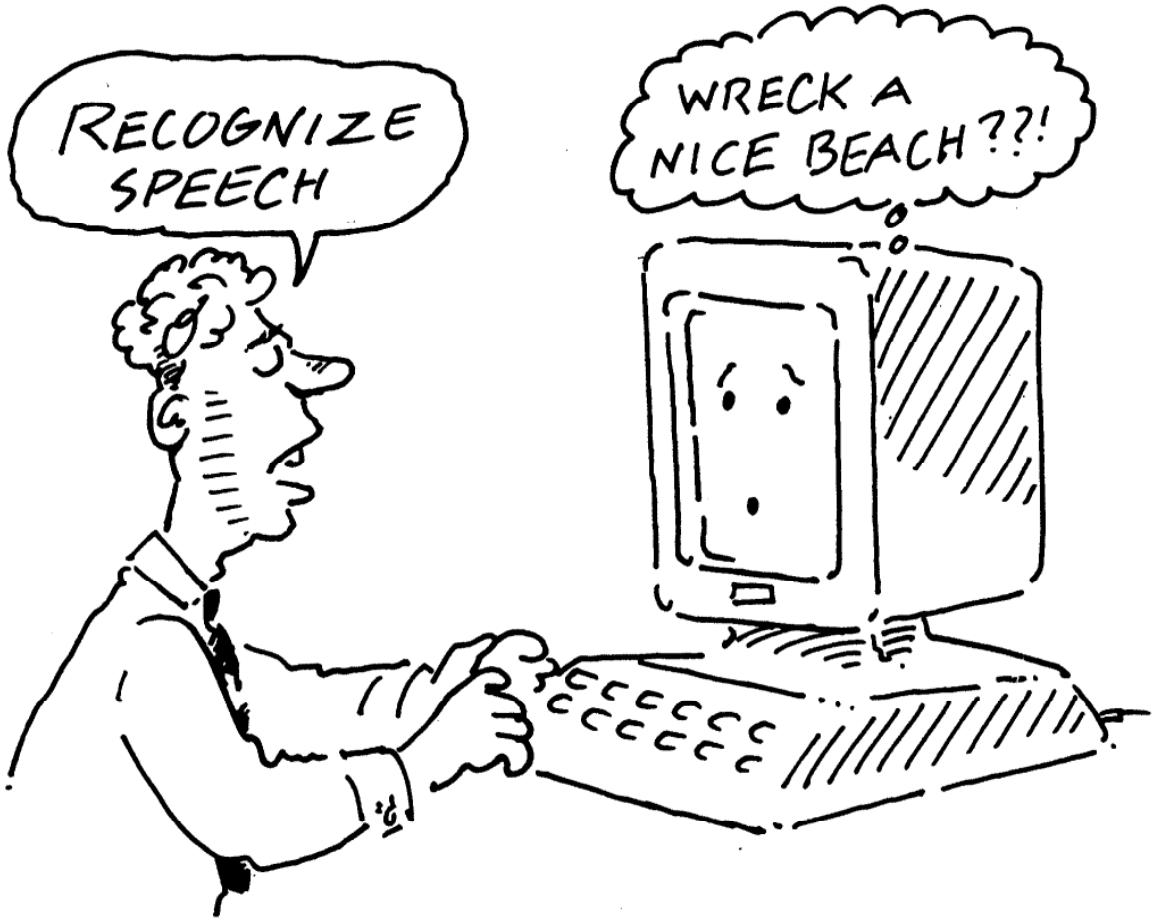
Voice

- Involves a person talking with a spoken language app
  - for example, search engine, train timetable, travel planner, or phone service or chatbots
- Used mainly for
  - inquiring about specific information, for example, flight times or the weather, or
  - issuing a command to a machine
    - such as asking a smart TV to select an action movie or asking a smart speaker to play some upbeat music
- Mostly a voice system is reactive responding to a person's queries
- Also used by people with visual impairments
  - For example, speech recognition word processors, page scanners, web readers, and home control systems

Voice

- The first generation of speech systems earned a reputation for mishearing all too often what a person said (see cartoon on the following page).
- However, they are now much more sophisticated and have higher levels of recognition accuracy.
- Machine learning algorithms have been developed that continue to improve their ability to recognize what someone is saying.

Have speech interfaces come of age?



Modeling human conversations

- People often interrupt each other in a conversation
  - Especially when ordering in a restaurant, rather than let the waiter go through all of the options
- Speech technology has a similar feature called 'barge-in'
  - Users can choose an option before the system has finished listing all of the options available

Barge-in feature in speech technolgoy

- barge-in feature allows callers to interrupt a system message and provide their request or response before the message has finished playing.
- This can be useful if the system has numerous options from which the caller may choose, and the chooser knows already what they want.

Structuring voice dialogs

- The most common way of structuring a dialogue is directed dialogue.
- Directed dialogs are where the system is in control of the conversation
  - Where it asks specific questions and requires specific responses
    - System: Which city do you want to fly to?
    - Caller: London
    - System: Which airport: Gatwick, Heathrow, Luton, Stansted, or City?
    - Caller: Gatwick
    - System: What day do you want to depart?
    - Caller: Monday next week.
    - System: Is that Monday, May 5?
    - Caller: Yes

Structuring voice dialogs

- More flexible systems allow the user to take the initiative:
  - For example, “I’d like to go to Paris next Monday for two weeks.”
  - More chance of error especially if the sentences are long
- Guided prompts can help callers back on track
  - For example, “Sorry I did not get all that. Did you say you wanted to fly next Monday?”

Voice assistants (for example, Alexa)

- Allow all to use rather than being single use
- Support families playing games, interactive storytelling, jokes, and so forth
- Can encourage social and emotional bonding
- Young children (under 4), however, find it difficult to be understood by the voice assistants
  - Frustrating for them

Research considerations

- How to design systems that can keep conversation on track
  - Help people navigate efficiently through a menu system enabling them to recover easily from errors
  - Guide those who are vague or ambiguous in their requests for information or services
- Type of voice actor (for example, male, female, neutral, or dialect)
  - Do people prefer to listen to and are more patient with a female or male voice, a northern or southern accent?

Pen-based devices

- Enable people to write, draw, select, and move objects at an interface using light pens or styluses
  - Capitalize on the well-honed drawing skills developed from childhood
- Smartpens use a combination of ordinary ink pen with digital camera that digitally records everything written with the pen on special paper

The LiveScribe Echo 2 smartpen

The Echo Smartpen records everything you write or draw on special Livescribe dot paper, capturing your notes and drawings in digital form.

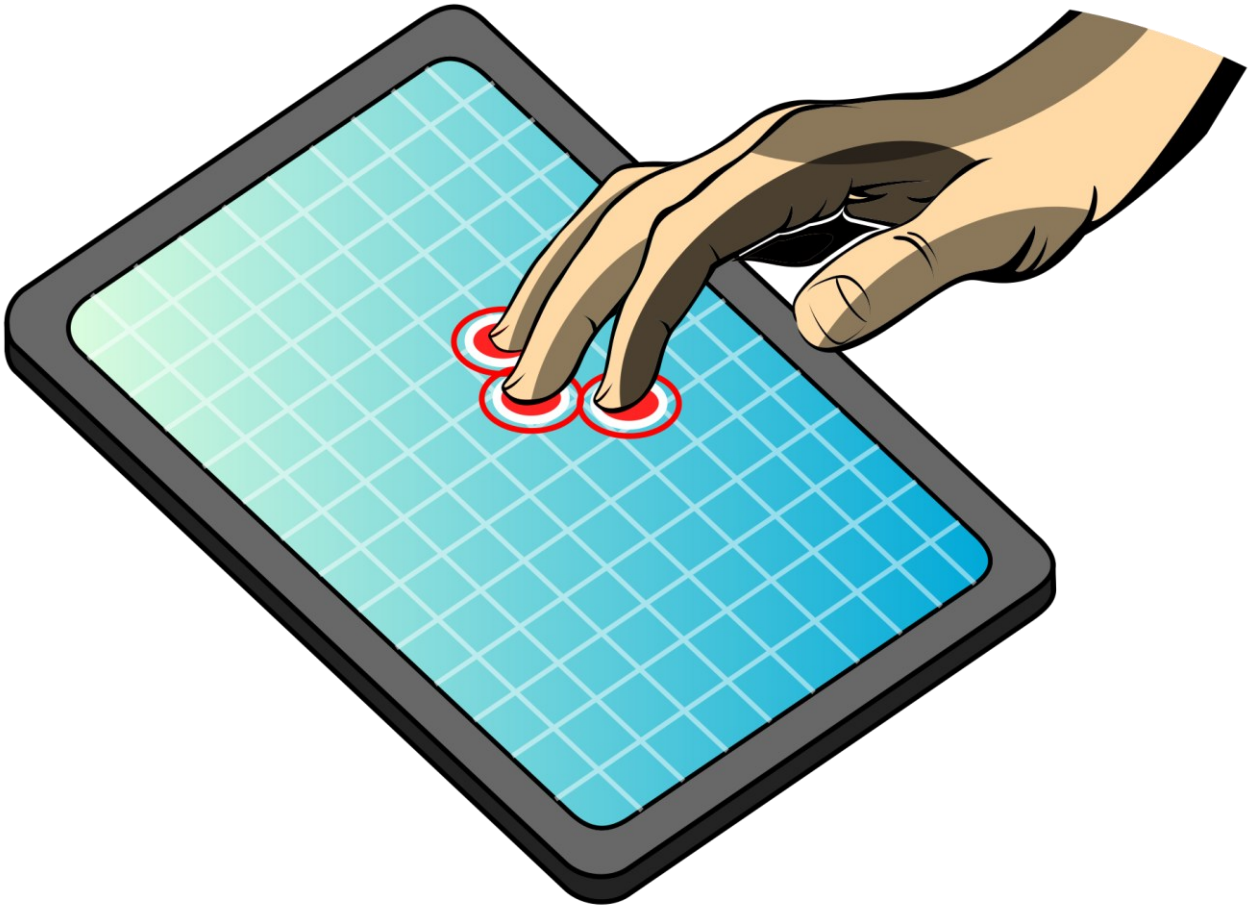
Advantages

- Allows people to annotate existing documents quickly and easily
- Can be used to fill in paper-based forms that can readily be converted to a digital record using standard typeface
- Can be used by remote teams to communicate and work on the same documents

Touchscreens

- Single touchscreens are used in walk-up kiosks (such as ticket machines and ATMs)
  - They work by detecting the presence and location of a person's touch on the display
- Multi-touch surfaces support a range of more dynamic finger tip actions, for example, swiping, flicking, pinching, pushing, and tapping
- They do so by registering touches at multiple locations using a grid
- Now used for many kinds of displays, such as smartphones, iPods, tablets, and tabletops
  - Supports one and two hand gestures, including tapping, zooming, stretching, flicking, and dragging

A multi-touch surface



Touchless

- Gestures involve moving arms and hands
  - to communicate (for instance, waving to say goodbye or raising an arm to speak in class) or
  - to provide information to someone (for example, holding two hands apart to show the size of something).

Touchless

- Uses camera, sensor, and computer vision techniques
  - Recognize people's arm and hand gestures in a room
  - Gestures need to be presented sequentially to be understood
    - in the same way a sentence is composed of a noun, then verb, and object plus operation.
    - For example, for "speaker, on," they used a gesture on one hand to designate the noun, and another on the other hand to designate the verb.

Touchless interface in the operating theater

- Surgeons need to keep their hands sterile during operations.
  - but also need to be able to look at X-rays and scans during an operation.
  - However, they need to avoid touching any keyboards, phones, and other nonsterile surfaces
- Microsoft's Kinect technology:
  - Surgeons could use gestures to interact with and manipulate MRI or CT images

Touchless interface in the operating theater

Recognizes core gestures for manipulating MRI or CT images using Microsoft Kinect

Car dashboards using gestures

- BMW's iDrive enables drivers to control certain functions with the use of hand gestures captured by a 3D camera
  - e.g. turning up or down the audio volume, For audio control these comprise of rotating the index finger clockwise (up) or anti-clockwise (down)
- Gesture control sensors are also positioned in the roof lining of the car
- To ensure they are safe, the gestures were developed and tested in the simulator

Touchless

- More recently, Touchless Computing has emerged as a new form of in-air gesture-based interaction that anyone can use with just a webcam
  - See www.intel.com/content/www/us/en/company-overview/wonderful/motion-input-technology.html

Research and design considerations (11)

- How are a user's gestures recognized and delineated?
  - Start and end points of a hand or arm movement ?
  - Difference between deictic and hand waving
- How realistic must the mirrored graphical representation (i.e., avatars) of the user be in order for them to be believable?

Haptic interfaces

- Provide tactile feedback
  - By applying vibration and forces (e.g., push) to a person's body, using actuators that are embedded in their clothing or a device they are carrying, such as a smartphone or smartwatch.
- For example, car steering wheels that are used with driving simulators can vibrate in various ways to provide the feel of the road
- The steering wheel in some cars like the Tesla also vibrate as a warning to the driver if the car detects that it is drifting out of its lane.

Haptic interfaces

- Vibrotactile feedback can be used to simulate the sense of touch between remote people who want to communicate
- Actuators embedded in clothing can be designed to recreate the sensation of a hug or a squeeze
- Ultrahaptics creates the illusion of touch in midair using ultrasound to make the illusion of 3D shapes and textures

Realtime vibrotactile feedback

- Another use of haptics is
  - to provide real-time feedback to guide people when learning a musical instrument, such as a violin or drums.
  - For example, the MusicJacket was developed to help novice violin players learn
    - how to hold their instrument correctly and
    - develop good bowing action

Realtime vibrotactile feedback

- Provides nudges when playing violin incorrectly
- Uses motion capture to sense arm movements that deviate from model
- Nudges are short vibrations on arms and hands

Research and design considerations (12)

- Where best to place actuators on body
- Whether to use single or sequence of ‘touches’
- When to activate, and how intense and how often to use them
- What kind of new smartphone/smartwatch apps can use vibrotactile creatively?
  - Apple iOS 16 (2022) recently added the option of being able to switch on their haptics when typing, enabling a user to feel a tiny vibration every time they press a key on their iPhone’s virtual keyboard

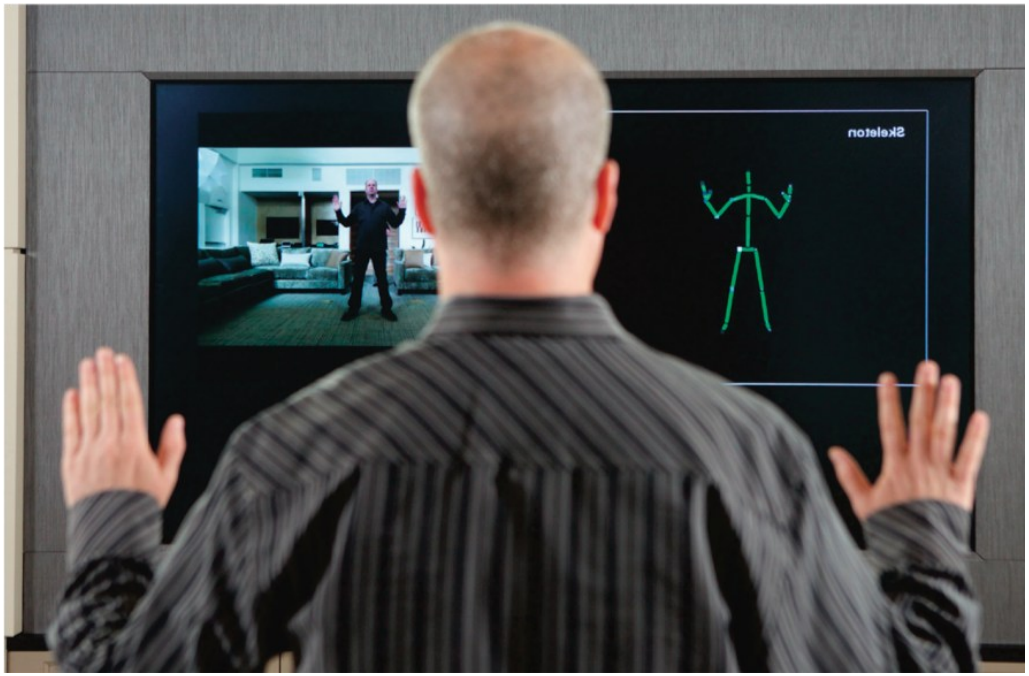
Multimodal Interfaces

- Provide enriched user experiences
  - By multiplying how information is experienced and detected using different modalities, such as touch, sight, sound, and speech
- Interface techniques that have been combined for this purpose include
  - speech and gesture, eye-gaze and gesture, haptic and audio output, and pen input and speech
- Most common combination is speech and vision

Multimodal Interfaces

- Support more flexible, efficient, and expressive means of human-computer interaction
- Can be combined with multi-sensor input to enable other aspects of the human body to be tracked
  - For example, eye gaze, facial expression, and lip movements
  - Provides input for customizing user interfaces to the user needs

Tracking a person's movements



- Kinect camera can detect multimodal input in real time using RGB camera for facial recognition and gestures, depth camera for movement tracking, and microphones for voice recognition
- Used to build graphical representation (avatar) of the user that could be programmed to move just like them

Research and design considerations

- Multimodal systems need to recognize and analyze user behavior, for example, speech, gesture, handwriting, or eye gaze
  - Much harder to calibrate these than single modality systems
- What is gained from combining different input and outputs
- Is talking and gesturing, as humans do with other humans, a natural way of interacting with a computer?

Shareable interfaces

Designed for more than one person to use:

- Provide multiple inputs and sometimes allow simultaneous input by co-located groups
- These include
  - large wall displays, for example, SmartBoards, where people use their pens or genstures
  - Interactive tabletops where small groups interact with information using their fingertips
    - For example, DiamondTouch, Smart Table, and Surface

A smartboard and an interactive tabletop interface

A SmartBoard in use during a meeting

Mitsubishi's interactive tabletop interface

Benefits (1)

- Provide a large interactional space that can support flexible group working
- Can be used by multiple users
  - Can point to and touch information being displayed
  - Simultaneously view the interactions and have the same shared point of reference as others
- Can support more equitable participation compared with groups using single PC

Augmented Reality (AR)

- Digital elemetns are superimposed on physical objects or the environment
- AR became a huge success with the arrival of Pokémon Go in 2016.
  - The game used smartphone camera and GPS to place virtual characters onto objects in the environment as if they really are there
- Many other applications including medicine, navigation, air traffic control, games, and everyday exploring

Other examples

In medicine

- Virtual objects, for example, x-rays and scans, are overlaid on part of a patient's body
- Aid the physician's understanding of what is being examined or operated

In air traffic control

- Dynamic information about aircraft overlaid on a video screen showing the real planes, and so on landing, taking off, and taxiing
- Helps identify planes difficult to make out

Augmented reality overlay on a car windshield

- A number of cars provide AR windshield technology, where navigation directions can literally look like they are painted on the road ahead of the driver
- In the following figure, weather alert with animated virtual lightning effects alongside information about a nearby café and the price of properties for sale or rent on a street.

Augmented reality overlay on a car windshield

Devices for viewing AR

- AR can be viewed through headsets, smartphones and glasses (e.g., SnapChat's AR spectacles).
- Headsets can provide a more immersive experience, but the downside is they are cumbersome to wear and fiddly to calibrate
- Convenient and easy-to-wear AR glasses are beginning to appear

Research and design considerations

- What kind of digital augmentation?
  - When and where in physical environment?
  - Needs to stand out but not distract from ongoing task
  - Needs to be able to align with real world objects
  - What happens if the AR is slightly off?
- What kind of device?
  - Smartphone, tablet, head up display or other?